

NETWORKING

Class-C Sub-netting

In this document we are going to see the complete process of sub-netting (Dividing a bigger network into small networks according to required number of hosts in a network) We will use FLSM (Fixed Length Subnet Mask) approach that will allocate equal number of ip's in each network. Let's have an IP address from class C **192.168.40.0**

Requirements:

- Needed 14 networks (Sub-nets)
- Each sub-net should contain at least **10** ips for hosts in each sub-net.

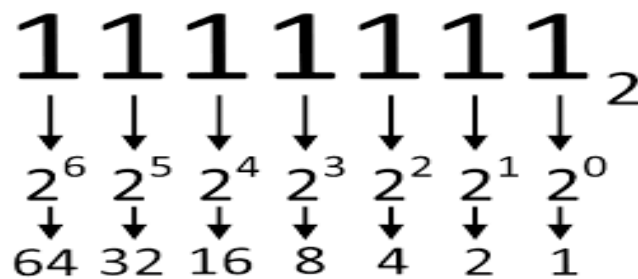
Solution:

As we are using class C so by default we know first three octets will represent **network** bits and last octet is for **hosts**. (**11111111. 11111111. 11111111.00000000**)

Network bits ==> 24

Host bits ==> 8

As per requirements we need to convert it into at least 14 sub-nets. So we can split last octet for required number of hosts(10 for each sub-net). Here is a last octet.



So here is the simple formula to see how many sub-nets we can get after borrowing bits from host part to network part (borrowing hosts bits from left to right.)

$$2^{\text{No. borrowing bits}} \geq \text{Req. number of sub-nets}$$

If we borrow three bits from host as,

11111111. 11111111. 11111111.11100000

$2^3 = 8$ (sub-nets) Which are less than we needed (14)

Now Borrowing 4 bits from host parts as,

11111111. 11111111. 11111111.11110000

$2^4 = 16$ (sub-nets). So according to mentioned formula above we can borrow 4 bits from host parts to fulfill need of 14 sub-nets. Now network bits becomes (24 + 4 = 28) which defines our sub-net mask (by turning on all borrowed bits. 128 + 64 + 32 + 16 = 240) Finally sub-net mask is

255.255.255.240 or /28

Calculating number of hosts in each sub-net (by viewing the value of last borrowed bit which is 2^4 (sequence number of last borrowed bit) = 16).

So after till now we come to know that we will have total **16 sub-nets** and each subnet will contain maximum **16 hosts** (1 Network IP, 1 broadcast IP and 14 usable host pi's).

First network will represent network IP/id and last IP broadcast IP. In between network and broadcast IP there are valid assignable host ip's.

We can get new sub-net by adding 16 to previous sub-net.

Note: We always start subnetting from start (0 for host bit of first sub-net)

Followings are details of each sub-nets

Sub-net	Network Address	Usable Host Range	Broadcast Address
1	192.168.10.0	192.168.10.1 - 192.168.10.14	192.168.10.15
2	192.168.10.16	192.168.10.17 - 192.168.10.30	192.168.10.31
3	192.168.10.32	192.168.10.33 - 192.168.10.46	192.168.10.47
4	192.168.10.48	192.168.10.49 - 192.168.10.62	192.168.10.63
5	192.168.10.64	192.168.10.65 - 192.168.10.78	192.168.10.79
6	192.168.10.80	192.168.10.81 - 192.168.10.94	192.168.10.95
7	192.168.10.96	192.168.10.97 - 192.168.10.110	192.168.10.111
8	192.168.10.112	192.168.10.113 - 192.168.10.126	192.168.10.127
9	192.168.10.128	192.168.10.129 - 192.168.10.142	192.168.10.143
10	192.168.10.144	192.168.10.145 - 192.168.10.158	192.168.10.159
11	192.168.10.160	192.168.10.161 - 192.168.10.174	192.168.10.175
12	192.168.10.176	192.168.10.177 - 192.168.10.190	192.168.10.191
13	192.168.10.192	192.168.10.193 - 192.168.10.206	192.168.10.207
14	192.168.10.208	192.168.10.209 - 192.168.10.222	192.168.10.223
15	192.168.10.224	192.168.10.225 - 192.168.10.246	192.168.10.247
16	192.168.10.248	192.168.10.249 - 192.168.10.254	192.168.10.255

~~~~~ Thank you 😊 ~~~~~